EXP NO:11 Experiment to understand pandas library use cases in Data science.

AIM:
To understand data frame use cases using the pandas library.

PROGRAM:

python

```python
import numpy as np
import pandas as pd

# Creating DataFrame using list
student_list = [
    [101, 'Arun', 85],
    [102, 'Rahul', 72]
]

df = pd.DataFrame(student_list)
df

# Assigning column names
df.columns = ['StudentID', 'Name', 'Marks']
df

# Display information
df.info()

# Reading CSV file
df = pd.read_csv("/content/student_performance.csv")
df

# First five rows
df.head()

# Last five rows
df.tail()

# Information about DataFrame
df.info()

# Selecting Marks column
df.Marks

# Checking type
type(df.Marks)

# Mean
df.Marks.mean()

# Median
df.Marks.median()

# Mode
df.Marks.mode()

# Variance
df.Marks.var()

# Standard deviation
df.Marks.std()

# Statistical description
df.describe()

# Statistical description including categorical data
df.describe(include='all')

# Getting column names
studentCol = df.columns
studentCol

# Converting DataFrame into NumPy array
studentarray = df.values
studentarray

# Creating DataFrame again from NumPy array
student_DF = pd.DataFrame(studentarray, columns=studentCol)
student_DF
```

OUTPUT:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2 entries, 0 to 1
Data columns (total 3 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   StudentID  2 non-null      int64
 1   Name       2 non-null      object
 2   Marks      2 non-null      int64
dtypes: int64(2), object(1)
memory usage: 180.0+ bytes
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 12 entries, 0 to 11
Data columns (total 3 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   StudentID  12 non-null     int64
 1   Name       12 non-null     object
 2   Marks      12 non-null     int64
dtypes: int64(2), object(1)
memory usage: 420.0+ bytes
```

	StudentID	Name	Marks
0	101	Arun	85
1	102	Rahul	72
2	103	Priya	91
3	104	Sneha	68
4	105	Vikram	77
5	106	Anjali	88
6	107	Karthik	54
7	108	Meera	95
8	109	Rohit	63
9	110	Divya	82
10	111	Suresh	72
11	112	Kavya	79

RESULT:

Thus the program to understand data frame use cases using the pandas library was executed successfully.